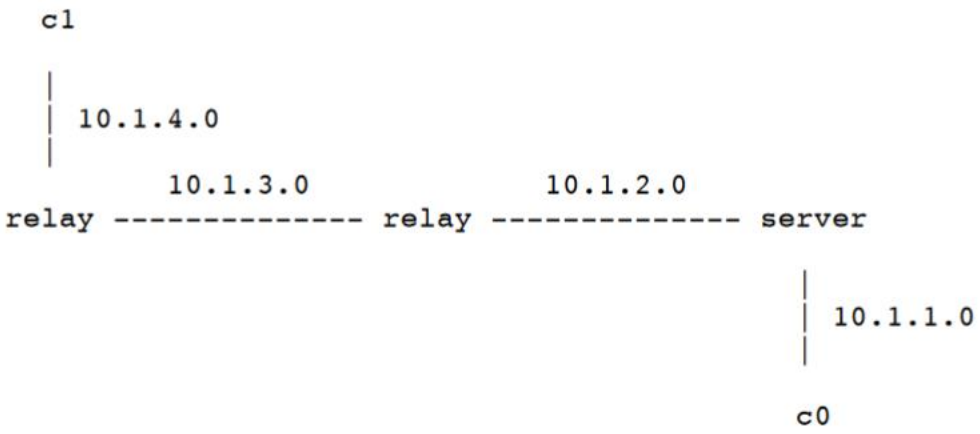
Félix Arnold

ECE 6110 - Lab Assignment 1
Topologies, Logging, and PCAP Tracing

Part 1a

In this first part, I modified the first.cc example from the ns3 tutorial to implement the following network topology, composed of two clients, a server and N relays. The number of relay can be set using a command line argument.



We can define the following parameters using the command line:

- the number of relays,
- the number of packets,
- the interval between two consecutive packets,
- the packet size,
- the data rate,
- the propagation delay.

Once created, we perform some tests on it.

The round-trip delays for client c1 is recorded with different parameter combinations. A computed round-trip delay, using a formula, is also added.

nRelays	packetSize (byte)	dataRate (Mbps)	Measured Round-trip Delay (s)	Computed Round-trip Delay (s)
1	256	5	0.00983	0.0096384
1	256	10	0.00892	0.0088192
1	1024	5	0.01475	0.0145536
1	1024	10	0.01137	0.0112768
2	256	5	0.01475	0.0144576

2	256	10	0.01337	0.0132288
2	1024	5	0.02212	0.0218304
2	1024	10	0.01706	0.0169152
3	256	5	0.01966	0.0192768
3	256	10	0.01783	0.0176384
3	1024	5	0.02949	0.0291072
3	1024	10	0.02275	0.0225536
4	256	5	0.02458	0.024096
4	256	10	0.02229	0.022048
4	1024	5	0.03686	0.036384
4	1024	10	0.02843	0.028192
5	256	5	0.02949	0.0289152
5	256	10	0.02675	0.0264576
5	1024	5	0.04424	0.0436608
5	1024	10	0.03412	0.0338304

To compute the round-trip delay, I used the following formula:

$$T_{round\ trip} = 2 \cdot (nRelays + 1) \cdot \left(\frac{8 \cdot packetSize}{dataRate \cdot 10^6} + delay \cdot 10^{-3}\right)$$

To explain this equation a little bit further, the round-trip time is the delay between the time the client c1 send its packet and the time it receives its echo. Let's break down the formula:

$\frac{8 \cdot packetSize}{dataRate \cdot 10^6}$ This part calculates the transmission time of a packet over a point-to-point connection. We need to take care to convert the values, notably the packet size in bytes into bits.

$delay \cdot 10^{-3}$ This is the propagation delay of a point-to-point connection.

By adding these two terms, we can compute the time a packet spends into each point-to-point connection.

$nRelays + 1$ This corresponds to the number of point-to-point connections between the client and the server.

However, we only get the delay of a one-way trip. Thus, we need to multiply all by 2.

Using this formula, the theorical values are very close to the one simulated. Yet, we still observe that they are slightly smaller than the measured round-trip time. This may be due to some delays in queuing or in processing, for example inside the server node when it needs to echo the packet.